AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M110
Python Programming

Final Examination
Spring Semester 2025/2026

Date: 09/Jun./2026

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Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 3 Hours

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are not allowed.

PART 1: Multiple Choices questions [10 Marks]

Q-1: Answer all the following questions by choosing the **most** correct statement.

1- A loop that stops when a special value is entered is called:
   a) Counter-controlled loop.
   b) Sentinel-controlled loop.
   c) Conditional loop.
   d) Indefinite loop.

2- Which of the following is a Boolean value in Python?
   a) 0.
   b) FALSE.
   c) True.
   d) All choices

3- Which of the following is used to handle exceptions in Python?
   a) handle
   b) try-except
   c) catch
   d) error

4- Which of the following is an invalid variable name?
   a) Value
   b) 2Value
   c) _Value
   d) All choices

5- Which function converts a tuple into a list?
   a) tuple().
   b) list_tuple.
   c) list().
   d) tuple_list.

6- Which of the following methods can be used to read a file line by line in Python?
   a) readln()
   b) readline()
   c) read()
   d) Both b and c

7- In Python, what does the 'self' keyword in a class refer to?
   a) The current class
   b) The parent class
   c) The instance of the class
   d) None of the above

8- In tkinter, which of the following methods starts the GUI event loop?
   a) start().
   b) run().
   c) self.root = Tk()
   d) mainloop().

9- What is the main purpose of a decision structure?
   a) To repeat statements.
   b) To execute statements in order.
   c) To terminate a program.
   d) To choose between alternative paths.

10- Which part of a function specifies its name and parameters?
   a) Body.
   b) Return statement.
   c) Header.
   d) Argument list.

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PART 2: Short Answer Questions [20 Marks]

You should dedicate approximately **70 minutes** to this part.

Question 2: [10 marks]
This question is composed of **four separate** parts; each is of weight equal to **2.5** marks.
Copy the complete code of each part to your answer booklet.

a- Complete the Python program below that is supposed to read student grades until -1 is entered, then calculate and print the average.

```
total_sum = 0
counter = 0
# Initial input
grade = float(input("Enter grade (or -1 to finish): "))
# The WHILE loop
_____:
   _____
   _____
   _____
# Calculate and print average
if counter > 0:
   average = total_sum / counter
   _____
else:
   print("No grades were entered.")
```

b- Complete the missing parts of the following Python program so that it:

- Asks the user to enter a string
- Ensures the string is not empty
- Prints the length of the string
- Prints its first character

```
_____=input("Enter a string: ")

while _____
print("Error: Empty string is not allowed.")
user_input = input("Enter a string: ")

print("Length:", _____)
print("First character:", _____)
```

c- Complete the following program to define a function named ***find_max*** that:

- Receives a list of integers as an argument.
- Returns the largest value in the list

The program should also print the largest value by calling the function.

```
def find_max(_____):
   max_value = numbers[0]
   for num in numbers:
      if _____ :
         _____
   return _____
nums = [4, 9, 2, 15, 6]
print(find_max(_____))   # Expected output: 15
```

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d- Complete the following program so that it prints the shape of asterisks (*)as shown below, based on the number entered by the user. If the user enters 4, the output should be:

**
***
****

```
n = int(input("Enter the number of rows: "))
for i in _____:
  for j in _____:
    print("*", end="")
  print()
```

Question 3: [10 marks]

1. Write a Python program that: **(4 marks)**
   a) Asks the user to enter a file name.
   b) Opens the file in ***write*** mode.
   c) Asks the user to enter 5 numbers.
   d) Writes each number to the file on a separate line.
   e) Closes the file properly after writing.

2. For the following list of tuples, where each tuple contains exactly three integers: **(6 marks)**
   my_list = [(2, 4, 6), (3, 5, 7), (1, 8, 4)]
   Write a complete Python program that:
   a) Uses a loop to calculate:
      - The sum of all first elements (2+3+1)
      - The sum of all second elements (4+5+8)
      - The sum of all third elements (6+7+4)
   b) Prints the three sums clearly.
   c) Then prints the average of each position.
   d) The program should work for any list of tuples of the same structure, not for only 3 tuples in the list.

PART 3: Programming/Problem Solving Questions [20 Marks]

You should dedicate approximately **70 minutes** to this part.

Question 4: [10 marks: 4+2+2+1+1]

You are requested to write a Python program that does the following:

a) Create a **Student** class with an initializer that takes **name**, **major**, and **gpa** as parameters.
b) Create an update_gpa() method to modify the **gpa** attribute by passing a new gpa as an argument.
c) Create an update_major() method to modify the **major** attribute by passing a new major as an argument.
d) Create a display() method that prints the current values of **name**, **major**, and **gpa**.
e) Testing the Class:
   - Instantiate the class with some initial values
   - Update the student's **gpa** and **major** using the methods
   - Display the updated student information using the display() method

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Expected Output (sample):

Initial Student Information:
Name: Noor
Major: CS
GPA: 3.2
Updated Student Information:
Name: Noor
Major: AI
GPA: 3.6

Question 5: [10 marks]

Consider the incomplete GUI program that should produce a window titled **"M110 GUI"** with:

- One label at the **top**: "Welcome to Tkinter"
- One label on the **left**: "Left Panel"
- One label on the **right**: "Right Panel"

You are asked to complete the program below (replace the dots with the proper code).

```
from tkinter import *

class M110_GUI:
   def __init__(self):
      self.root = ......
      .........
      self.label_top = .........
      self.label_left = .........
      self.label_right = .........
      .........
      .........
      .........
      .........

app = .........
```

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End of Questions